ALMA Proposal Preparation Workshop

March 25, 2026 @INCT-UDA, Copiapó

2.3. ALMA Proposal Preparation

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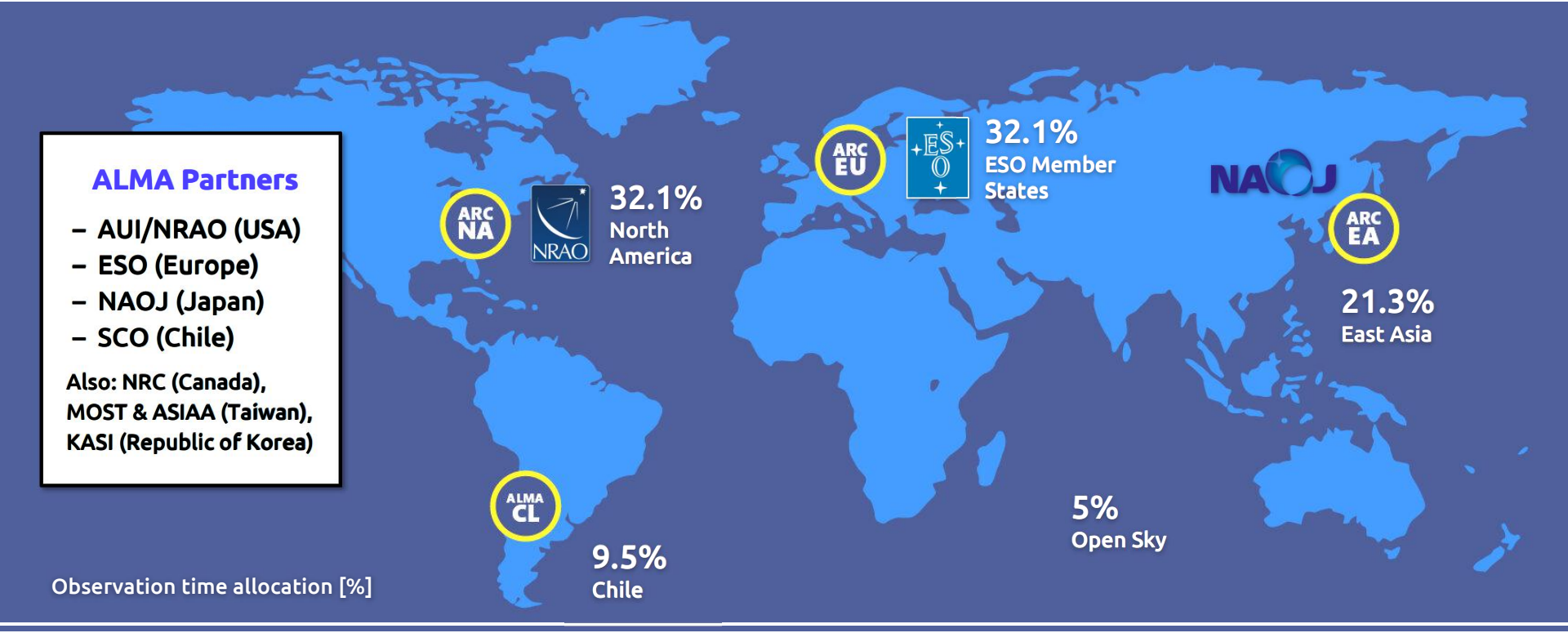
Partners & Time Allocation



Starter Checklist

Do you have a great proposal idea?

- ☑ Target observability
- ☑ ALMA Science Archive
- ☑ Observing parameters
- ☑ Technical feasibility (OT)
- ☑ Ancillary data (e.g., APEX)

Observing Strategy

Proposal Preparation Overview

CfP deadline: April 23

- **Create an ALMA account**
  https://asa.alma.cl/UserRegistration/newAccount.jsp?arc=eu
  - Permanent username
  - Select expertise keywords
- **Observing Tool (OT)**
  https://cycle-13.sps.alma.cl/ngot/
  - Design the Science Goals in OT, test project feasibility
- **Project preparation**
  - Prepare the science case (SJ; must be strictly anonymous)
  - Justify technical setup (TJ)

Timeline

Key considerations

→ Sensitivity required
→ Angular resolution
→ Spectral resolution
→ Target size scales

Range of angular scales

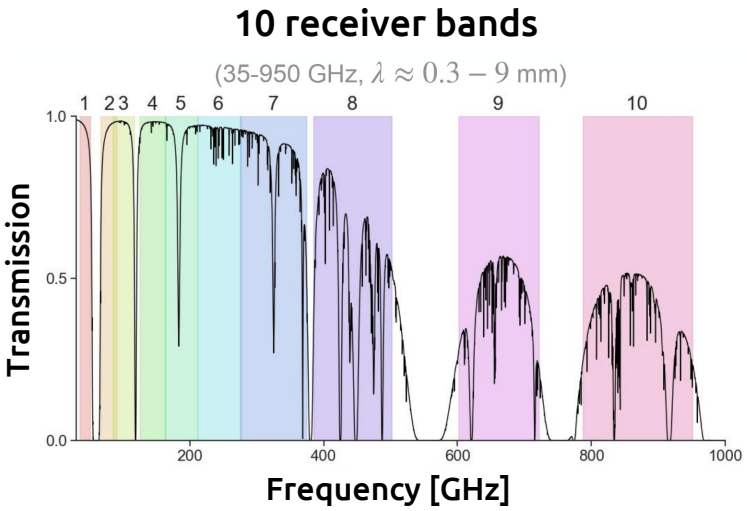
$$\lambda/B_{max} < \theta < \lambda/B_{min}$$

$$AR < \theta < MRS$$

AR: Angular Resolution
MRS: Maximum Recoverable Scale

Observing Strategy: Receiver Bands

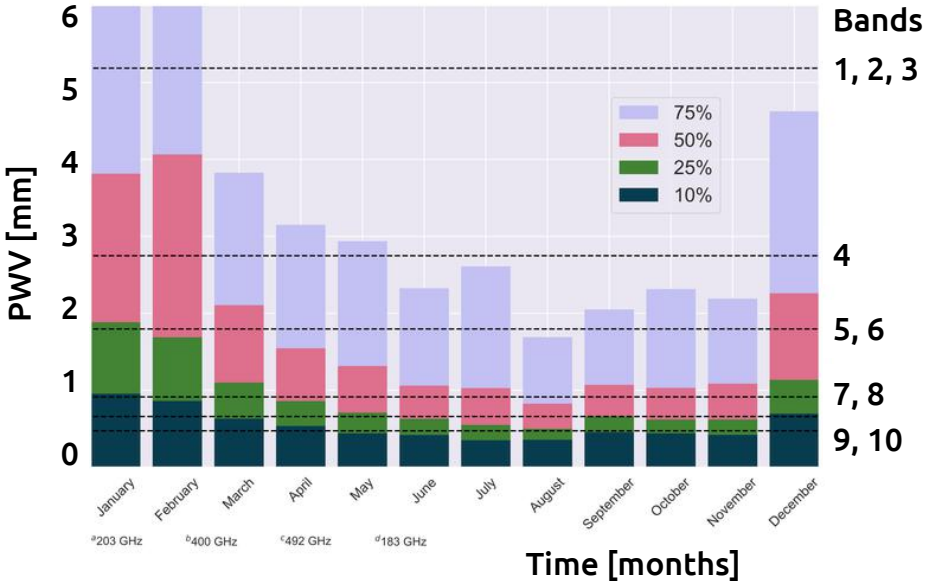
Band	Frequency range [GHz]	Wavelength range [mm]	Type
1	35 – 50	8.5 – 6	SSB
2	67 – 116	4.5 – 2.6	2SB
3	84 – 116	3.6 – 2.6	2SB
4	125 – 163	2.4 – 1.8	2SB
5	158 – 211	1.9 – 1.4	2SB
6	211 – 275	1.4 – 1.1	2SB
7	275 – 373	1.1 – 0.8	2SB
8	385 – 500	0.78 – 0.60	2SB
9	602 – 720	0.50 – 0.42	DSB
10	787 – 950	0.38 – 0.32	DSB



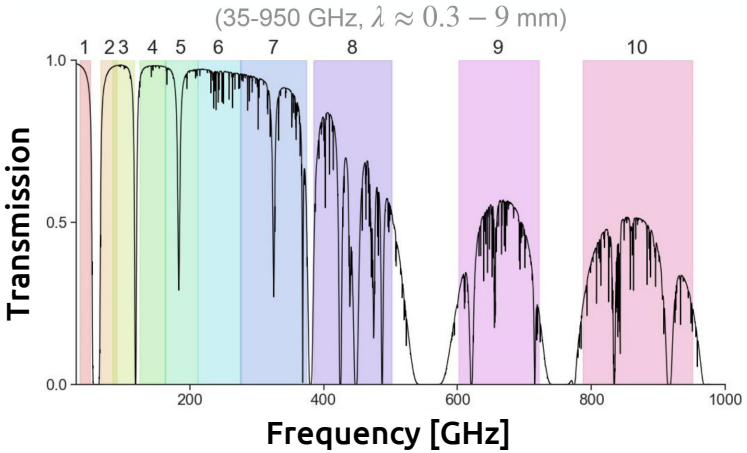
Cycle 13 Proposer's Guide: https://almascience.nrao.edu/proposing/proposers-guide

Observing Strategy: Weather Conditions

Fraction of time that the PWV falls below a given PWV value over the year



10 receiver bands



Cycle 13 Proposer's Guide: https://almascience.nrao.edu/proposing/proposers-guide

Cycle 13 Array Configurations

Start date	Configuration	Max Baseline
1 Oct 2026	C-3	0.50 km
20 Oct 2026	C-2	0.31 km
10 Nov 2026	C-1	0.16 km
30 Nov 2026	C-2	0.31 km
20 Dec 2026	C-3	0.50 km
10 Jan 2027	C-4	0.78 km
1 Feb 2027	Maintenance	Maintenance
1 Mar 2027	C-4	0.78 km
20 Mar 2027	C-5	1.4 km
20 Apr 2027	C-6	2.5 km
20 May 2027	C-7	3.6 km
20 Jun 2027	C-8	8.5 km
11 Jul 2027	C-9	13.9 km
30 Jul 2027	C-10	16.2 km
20 Aug 2027	C-9	13.9 km
10 Sep 2027	C-8	8.5 km

ALMA antenna transporter

Credit: NRAO

One day to move an antenna

- Configurations: **C-1 to C-10**
- Maximum baselines: from **0.16 km** to **16 km**

https://almascience.nrao.edu/observing/observing-configuration-schedule/long-term-configuration-schedule

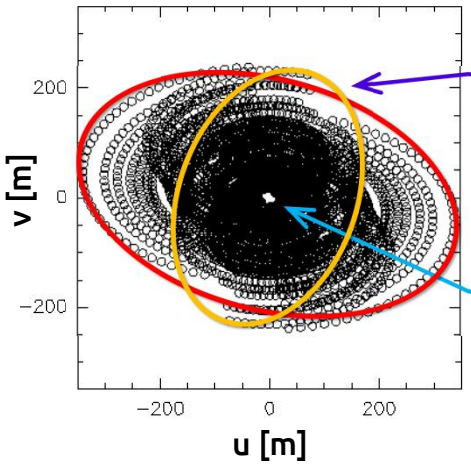
- **Highly encouraged projects**
  - Long baselines (C-7 to C-10) at lower frequencies (Bands 1–4)
  - High-frequency (Bands 8–10)
  - Stand-alone ACA (= no 12-m)

https://almascience.nrao.edu/proposing/proposers-guide

Configurations & Angular Scales

Start date	Configuration	Max Baseline
1 Oct 2026	C-3	0.50 km
20 Oct 2026	C-2	0.31 km
10 Nov 2026	C-1	0.16 km
30 Nov 2026	C-2	0.31 km
20 Dec 2026	C-3	0.50 km
10 Jan 2027	C-4	0.78 km
1 Feb 2027	Maintenance	Maintenance
1 Mar 2027	C-4	0.78 km
20 Mar 2027	C-5	1.4 km
20 Apr 2027	C-6	2.5 km
20 May 2027	C-7	3.6 km
20 Jun 2027	C-8	8.5 km
11 Jul 2027	C-9	13.9 km
30 Jul 2027	C-10	16.2 km
20 Aug 2027	C-9	13.9 km
10 Sep 2027	C-8	8.5 km

uv-plane sampling



Outer boundary

Sets the angular resolution limit:

$\theta_{res} = \lambda/B_{max}$

Inner hole

Sets the MRS:

$\theta_{MRS} = \lambda/B_{min}$

Synthesized beam (θ_{res})

Configurations & Angular Scales

➤ See Tables A-1 (θ_{min}) and A-2 (θ_{MRS}): https://almascience.nrao.edu/proposing/proposers-guide#autotoc-item-autotoc-69

Config.	Band 1 @40 GHz	Band 2 @75 GHz	Band 3 @100 GHz	Band 4 @150 GHz	Band 5 @185 GHz	Band 6 @230 GHz	Band 7 @345 GHz	Band 8 @460 GHz	Band 9 @650 GHz	Band 10 @870 GHz
7-m	θ_{res} = 31.8″ θ_{MRS} = 167″	16.9″ 88.9″	12.7″ 66.7″	8.47″ 44.5″	6.87″ 36.1″	5.52″ 29.0″	3.68″ 19.3″	2.76″ 14.5″	1.95″ 10.3″	1.46″ 7.67″
C-1	θ_{res} = 7.69″ θ_{MRS} = 71.2″	4.10″ 38.0″	3.07″ 28.5″	2.05″ 19.0″	1.66″ 15.4″	1.34″ 12.4″	0.89″ 8.25″	0.67″ 6.19″	0.47″ 4.38″	0.35″ 3.27″
C-2	θ_{res} = 7.31″ θ_{MRS} = 56.5″	3.89″ 30.1″	2.92″ 22.6″	1.95″ 15.0″	1.58″ 12.2″	1.27″ 9.81″	0.85″ 6.54″	0.64″ 4.90″	0.45″ 3.47″	0.34″ 2.59″
C-3	θ_{res} = 4.79″ θ_{MRS} = 40.5″	2.55″ 21.6″	1.91″ 16.2″	1.28″ 10.8″	1.03″ 8.73″	0.83″ 7.02″	0.55″ 4.68″	0.42″ 3.51″	0.29″ 2.48″	0.22″ 1.86″
C-4	θ_{res} = 2.99″ θ_{MRS} = 28.0″	1.60″ 14.9″	1.20″ 11.2″	0.80″ 7.50″	0.65″ 6.08″	0.52″ 4.89″	0.55″ 3.26″	0.26″ 2.44″	0.18″ 1.73″	0.14″ 1.29″
C-5	θ_{res} = 1.80″ θ_{MRS} = 16.8″	0.96″ 8.93″	0.72″ 6.70″	0.48″ 4.47″	0.39″ 3.62″	0.31″ 2.91″	0.21″ 1.94″	0.16″ 1.46″	0.11″ 1.03″	0.083″ 0.77″

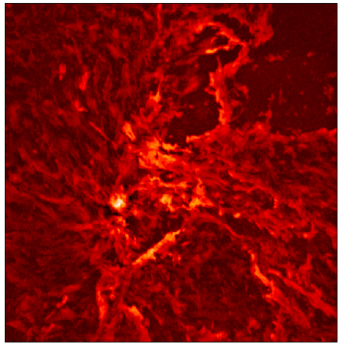
➤ **Example:** I would like to image at ~2″ resolution the CO(1–0) line at 46.5 GHz (Band 1) in a ~20″-extended lensed source at z = 1.5.

Using C-4: θ_{res} = 2.99″ × (40/46.5) = 2.57″ and θ_{MRS} = 28.0 × (40/46.5) = 24.1″ > 20″ ☺

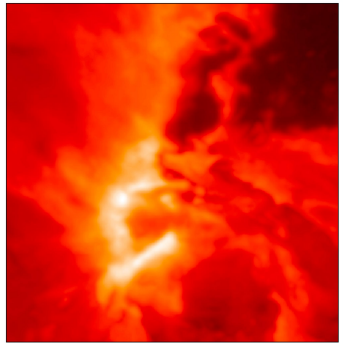
Using C-5: θ_{res} = 1.80″ × (40/46.5) = 1.55″ and θ_{MRS} = 16.8 × (40/46.5) = 14.5″ < 20″ ☹

Configurations & Angular Scales

What if the target > MRS? → + C-(N-x) or 7-m



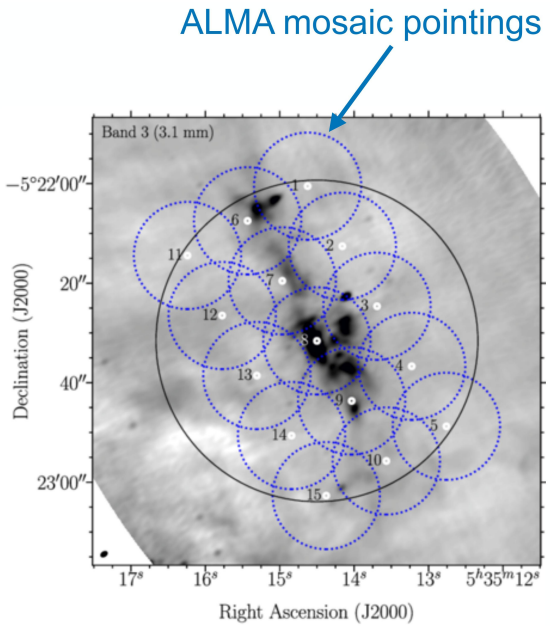
High resolution



Lower resolution

➤ The Observing Tool will automatically select a less extended array configuration (i.e., with larger θ_{res} and larger θ_{MRS}) to recover flux on larger scales.

What if the target < Primary Beam? → Mosaic



Primary Beam ≡ FOV ~ λ/D

ALMA Cycle 13 Capabilities

Summary

- CfP: **March 19** to **April 23** (5 weeks)
- All receiver bands will be offered, including new **Band 2** (for the 12-m array only)
- The OT choice between Band 2 and 3 will be described in the Cycle-13 Proposer's Guide
- Largest configurations **C-9** & **C-10** offered
- Cycle 13 is not impacted by the WSU
- New web-based Observing Tool version (for the Phase-1 submissions only)

Sources of Support

- ALMA Helpdesk

  Questions are usually answered within 48 hours
  Submit a ticket: https://help.almascience.org

- Face-to-face Visitor Support

  The NRAO/NAASC offers face-to-face meetings (online or in-person) for proposal preparation support and data reduction (PI data or archival)
  https://science.nrao.edu/facilities/alma/visitors-shortterm

Proposal Writing

Program Category

ALMA Proposal Types

- **Regular Program** (PID 2026.1.xxx.S)
  Time limit: ≤ 50 h (12-m Array)
  ≤ 150 h (7-m stand-alone)
- **Very Long Baseline Interferometry (VLBI)**
  ALMA used as part of a global VLBI network
- **Phased Array**
  Antennas phased together as a single dish
- **Target of Opportunity (ToO)**
  Triggered by unpredictable/transient events
- **Large Program** (PID 2026.1.xxx.L)
  Major, legacy-type projects
  > 50 h (12-m) or > 150 h (7-m)

Joint Proposals

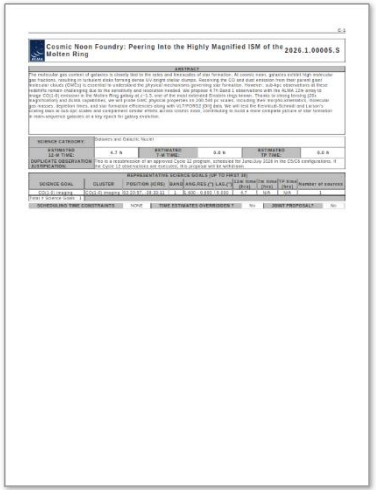
Facility	Available Time
JWST	115 h /cycle
VLA	50 h /year
VLT	50 h /year

Two or more observatories are required to achieve the science

- ALMA has agreements with JWST, VLA and VLT
- Science review by the **main** observatory plus technical review by all observatories

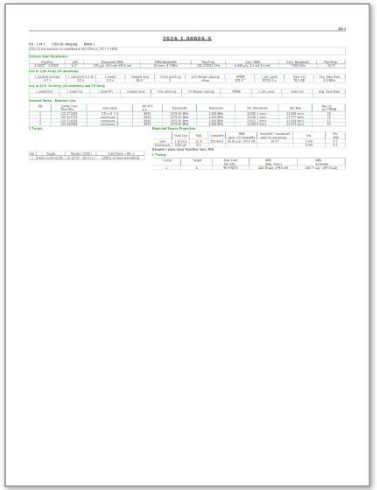
Regular Programs

➤ See Proposal Writing slides



Proposal title, abstract, total time, SG list

Science Case (max 4 pages)



Science Goal summary

Technical Justification